

TAMER ELKOT

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on expertise in Computer Vision, MLOps, and Backend Development. Experienced in designing and deploying end-to-end AI systems — from data pipelines to production inference servers — using FastAPI, Docker, TensorFlow, and PostgreSQL. Proven ability to lead cross-functional teams and deliver measurable results in real-world environments. Passionate about applying AI to solve impactful problems in agriculture, logistics, healthcare, and security.

TECHNICAL SKILLS

Languages	Python, C, SQL (PostgreSQL)
AI / CV	TensorFlow, Keras, OpenCV, Transformers, Vision Language Models (VLMs), Scikit-learn, ML Models, Transfer Learning, YOLO, NLP Basics.
Backend	FastAPI, REST APIs, SQLAlchemy, OOP, Streamlit, Django. Docker, Linux (Ubuntu), Git, Celery.
Tools	VS Code, Jupyter, Google Colab, Hugging Face, Kaggle, Claude Code, Earth Engine APIs, Roboflow.

PROFESSIONAL EXPERIENCE

AI Engineer Intern | [SEG](#) Apr 2026 – Present

- **Cost Optimization & Profitability:** Developing and deploying B2B AI solutions focused on reducing operational overhead and maximizing product profit margins through predictive modeling.
- **Business Intelligence Integration:** Collaborating with stakeholders to translate complex data insights into actionable B2B strategies, directly impacting the product's bottom line.

AI Mentor | [Celulla Technology](#) Apr 2026 – Present

- **Technical Mentorship & Code Quality:** Lead and mentor a cohort of AI interns by conducting rigorous line-by-line code reviews, ensuring adherence to PEP 8 standards, modular programming, and optimized model performance.
- **Curriculum Development & Technical Support:** Develop and publish specialized academic content covering advanced AI concepts while providing real-time technical troubleshooting to resolve blockers in machine learning pipelines.
- **Project Oversight & Agile Collaboration:** Manage weekly task allocation and progress tracking through sync meetings, facilitating technical discussions to overcome algorithmic challenges and ensure timely project milestones.

Computer Vision Engineer Intern | [Celulla Technology](#) Jan 2026 – Mar 2026

- **Project Lead – Shoplifting Detection:** Directed a team to develop a real-time video classification system using CNN + LSTM (LRCN) and MoViNet architectures to distinguish suspicious from normal behavior.
- **Water Segmentation (Satellite Imagery):** Engineered an AI Water Sentinel tool using Sentinel-2 (12-channel) data and a custom U-Net for global flood monitoring; project was featured on the company's official LinkedIn page.

Teeth Classification: Built an end-to-end dental diagnostic system using FastAPI and Streamlit, supporting real-time switching between a Custom CNN and EfficientNetB0 for multi-class tooth identification from X-rays

Machine Learning Engineer Intern | Technocolabs Softwares Jan 2026 – Present

- **Predictive Risk Engine:** Building an end-to-end SLA breach predictor for logistics.
- **Data Analysis:** Analyzed 200,000+ shipment records; identified Fragility (+14% risk) and Temperature Control (+10% risk) as primary failure drivers.
- **MLOps Pipeline:** Engineered a robust pipeline with data validation and leakage prevention for XGBoost modeling, managed via ..

PROJECTS

AgriVision | U-Net · Sentinel-2 12-Channel · FastAPI · Docker · PostgreSQL Mar 2026

- **Geospatial Pipeline:** Engineered a robust 12-channel multispectral data pipeline using Sentinel-2 imagery with advanced radiometric normalization and augmentation to isolate water signatures across diverse global terrains.
- **Deep Learning Architecture:** Trained a custom U-Net for semantic segmentation with a hybrid Dice + Binary Cross-Entropy loss to mitigate class imbalance and achieve high-precision pixel-level accuracy.
- **MLOps Integration:** Deployed via FastAPI for asynchronous processing and Docker for containerization, ensuring seamless model inference and pipeline portability.
- **Temporal Analytics:** Built an interactive geospatial dashboard for real-time ROI selection and historical tracking, persisting time-series water data in PostgreSQL for agricultural and flood analysis.

EmpVision | YOLOv8 · FaceNet · FastAPI · Docker · PostgreSQL Feb 2026

- **Biometric System:** Replaced legacy HOG methods with YOLOv8 + FaceNet, achieving high accuracy at 90° angles and in low-light conditions.
- **Backend Architecture:** Designed a FastAPI + PostgreSQL backend using native ARRAY types to store 512-dimensional face embeddings for vector similarity search.
- **Real-Time Streaming:** Designed a multi-threaded RTSP pipeline for zero-latency streaming with Passive Video Enrollment to filter blurred frames.

Employee Turnover Prediction | XGBoost · SMOTE · Streamlit · FastAPI · Docker Aug 2025

- **Model:** Achieved 99% prediction accuracy using a tuned XGBoost model with SMOTE for class imbalance handling.
- **Deployment:** Dockerized and deployed via Streamlit, making the predictive tool accessible for end-users.

Depression Rate Survey | XGBoost · Optuna · Stratified K-Fold — Kaggle Top 8 of 500+ Jan 2025

- **Achievement:** Built an ML pipeline achieving 92.09% accuracy, ranking Top 8 of 500+ participants using Optuna hyperparameter tuning and Stratified K-Fold Cross-Validation.

EDUCATION

Bachelor of Science in Mechatronics Engineering | Higher Institute of Engineering & Technology |
Expected Graduation: May 2027

CERTIFICATIONS

- HCIA-AI V4, Huawei ICT Academy – Jul 2025
- Machine Learning & Deep Learning, NTI – Aug 2025
- Deep Learning for Computer Vision, NTI – Sep 2025
- Artificial Intelligence, NTI – Jul 2025
- Machine Learning & Deep Learning, Neurotech – Apr 2025